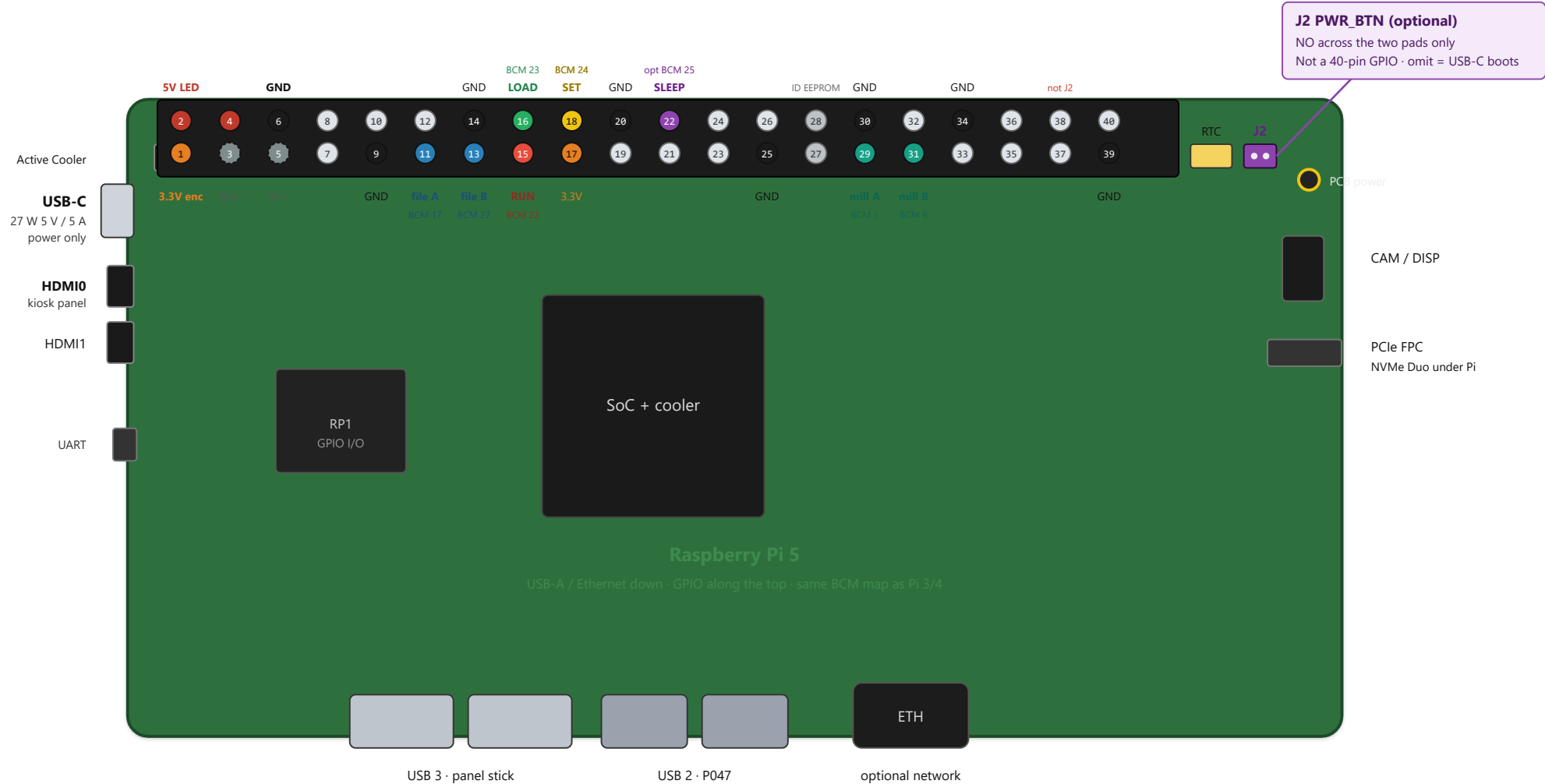


fh6parse 1.4.0 — Raspberry Pi 5 kiosk wiring

Hardware sheet only. Software / OS / UPDATE: packaging/LINUX-KIOSK.md. GPIO is **3.3 V**. BCM numbers, not header pin numbers. Optional parts may stay unwired. **Page 4** is the screw terminal, wire gauge and colour card. NVMe Base Duo under the Pi. Panel USB 3.0 and Active Cooler are not GPIO.

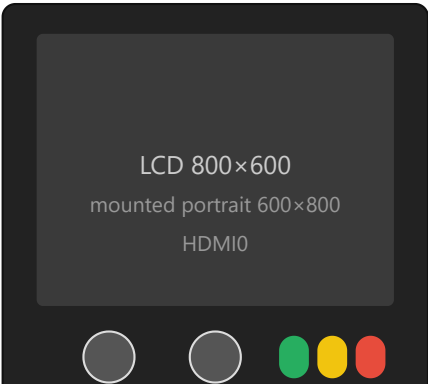
40-pin GPIO · pin 1 at USB-C end · outer row = even (board edge) · inner row = odd (SoC)



3.3 V (encoders) 5 V (LED rings only) GND (common) File EC11 Mill EC11 LOAD green SET yellow RUN red Sleep (optional) I²C1 reserved* Unused

* Pins 3 / 5 (BCM 2 / 3, I²C1 SDA / SCL) are left free for a future Qwiic keypad. Do not steal them for F2 or extra buttons. Grey dashed = reserved, not required. Pin 38 (BCM 20) is **not** the power button.

1. Front panel (as you face the kiosk)



Buttons under the screen, left → right



header 16
BCM 23



header 18
BCM 24



header 15
BCM 22



header 22
BCM 25

Leave SLEEP unwired: pin sits on pull-up, never
POWER is J2, not this GPIO.

2. DFRobot Fermion EC11 (SEN0235)

Two knobs, same module. 3.3–5 V — power from **3.3 V only**. Shaft click **C unused**.

EC11 silk	File knob	Mill knob
VCC	pin 1 (3.3 V)	pin 1 or 17 (3.3 V)
GND	pin 6 (GND)	pin 30 (GND)
A (CLK)	pin 11 · BCM 17	pin 29 · BCM 5
B (DT)	pin 13 · BCM 27	pin 31 · BCM 6
C (push)	not connected	

Wrong direction → settings **Reverse** (or swap A/B). 20 pulses/turn; start **Ticks per tooth = 2** if one click skips two names.

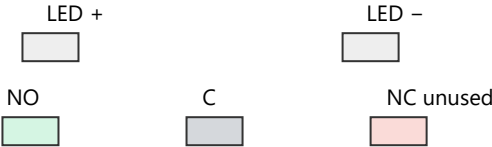
4. Optional J2 power (not GPIO)

Purple 2-pad header next to the RTC battery, GPIO pin-40 corner. Same switch as the PCB button.

Do	Do not
Momentary NO across the two J2 pads	Tie J2 to BCM 25, BCM 20, or GPIO GND
Leave J2 open if unused — USB-C still boots	Expect Pi 4 WAKE_ON_GPIO on GPIO3
Keep the PCB button reachable	LED rings on J2 · WAIT_FOR_POWER_BUTTON without J2

3. 16 mm 5-pin vandal (LOAD / SET / RUN / optional SLEEP)

Switch and ring LED are separate. Meter the tabs — clone silk varies. Rear view (typical):



Tab	To Pi
NO	GPIO (pull-up). Press = GND
C	common GND (pin 6 / 9 / 14 / 20 / 30)
NC	leave open
LED +	5 V pin 2 (or 4). Not 3.3 V
LED -	GND

5. Power and USB (not GPIO)

Port	Use
USB-C (Pi)	Official 27 W PSU. Power only
micro-HDMI0	Kiosk LCD (port next to USB-C)
USB / FAN	Panel 27 mm USB 3.0 stick → Pi USB 3. P047 → USB 2. Official Active Cooler on FAN
PCIe FPC	Pimoroni NVMe Base Duo under the Pi. Official Pi 512 GB 2230 in slot A. Not the kit M.2 HAT+
Ethernet	Optional (git / UPDATE / company STEP share)

Do not power the P047 from Pi USB. Rings stay on 5 V after halt unless you switch that 5 V.

6. Default map (header pin → BCM → function)

Header	BCM	Function	Required
1	—	3.3 V · encoder VCC	yes if knobs fitted
2 / 4	—	5 V · vandal LED +	rings only
3	2	I ² C1 SDA — reserved, leave open	no
5	3	I ² C1 SCL — reserved, leave open	no
6, 9, 14, 20, 30	—	GND · common for knobs, switches, LED –	yes
11	17	File encoder A (CLK)	yes
13	27	File encoder B (DT)	yes
15	22	RUN red NO	yes
16	23	LOAD green NO	yes
17	—	3.3 V spare (same rail as pin 1)	no
18	24	SET yellow NO	yes
22	25	SLEEP / wake NO	no — unconnected is fine
29	5	Mill encoder A (CLK)	yes
31	6	Mill encoder B (DT)	yes
J2 pads	—	Panel POWER (parallel with PCB button)	no — USB-C still boots

Change BCM numbers in kiosk settings (**F2**) or `/etc/fh6parse-kiosk.ini`. Overlay: `~/ .config/fh6parse/ui.ini`. Land these on the **52Pi screw terminal** (same header numbers). Do not also push Dupont onto the 40-pin under the board.

7. Common ground

One GND net for encoders, every C tab, every LED –, and (if fitted) a Qwiic keypad. Tie them to any GND header pin listed above. Do not run a separate “button ground” to J2.

8. Identify vandal tabs with a meter

- 1. Diode pair = LED + / – (one way ~1.8–3 V).
- 2. Unpressed: C–NC beeps, C–NO open.
- 3. Pressed: C–NO beeps. The tab that beeps in both tests is **C**.

Do not

- Put 5 V on A / B / C / NO / NC or any BCM pin
- Power EC11 VCC from pin 2 (5 V) unless a shifter is on A/B
- Wire J2 to GPIO 25 or GPIO 20
- Use `RPi.GPIO` on Pi 5 (RP1 needs `Igpio`)
- Set `WAIT_FOR_POWER_BUTTON=1` without a J2 (or reachable PCB) button

Without optional parts the kiosk still runs: idle blanks the panel; knobs / USB / keyboard wake it; USB-C plug-in boots the Pi. gpiozero pull-ups keep unused BCM 25 high.

9. Wires (52Pi screw terminal)

Pi GPIO end is the **screw terminal board** ([Kamami 588019](#)). Same 40-pin header as Pi 4B; pin 1 at USB-C. Brass M2.5 standoffs in the kit. If it will not clear the Active Cooler / FAN plug, sit it on a **40-pin GPIO riser** (same pin 1). Silk I017 = BCM 17. Leave header 3 / 5 empty (Qwiic). **J2 is not on this board**. GPIO sense, LED rings, and Pi power must never share a conductor that can put 5 V on a BCM pin. No jumper-wire packs.

Buy

Circuit	Wire	Ends
Encoder A/B, button NO (3.3 V sense)	24 AWG stranded (0.25 mm²)	Screw on 52Pi; Dupont/solder at EC11
Encoder VCC 3.3 V , common GND	22 AWG stranded (0.34 mm²)	Screw on 3.3 V (pin 1 / 17) and GND
LED + 5 V and LED –	22 AWG stranded	Screw 5 V pin 2 / GND; 2.8 mm Faston on the vandal
Optional J2 power	26–28 AWG , keep < 20 cm	J2 pads only — not a GPIO screw
Future Qwiic keypad	Official Qwiic 4-pin cable	Do not steal SDA/SCL screws

24 AWG GPIO under ~40 cm. **22 AWG** for 3.3 V / 5 V / GND. Terminals ~16–26 AWG; stay 22/24. Stranded. Ferrule on the screw; do not tin a blob. Silicone next to the Pi 5 cooler.

Terminations

- **Pi**: screw terminal on the 40-pin header, or on a GPIO riser if the cooler is in the way. Brass standoffs. Strip ~5 mm, ferrule. Strain-relieve to the case, not the board.
- **Vandal**: insulated 2.8 mm Faston females, **crimped**. Do not solder on the switch (the plastic melts). Leave **NC** empty.
- **EC11**: Dupont or solder onto the pin row. Leave shaft **C** open.
- **J2**: two short flying leads across the pads only. Never a screw on the GPIO block.

Colour

Colour	Net
Black	GND (knobs, C tabs, LED –)
Orange	3.3 V encoder VCC (pin 1 / 17)
Red	5 V LED + only (pin 2)
Blue / light blue	File encoder A / B
White / grey	Mill encoder A / B
Green	LOAD NO (BCM 23)
Yellow	SET NO (BCM 24)
Brown	RUN NO (BCM 22) — not red
Violet	SLEEP NO (BCM 25), if fitted
Grey twisted pair	J2 only

Do not use the same red for RUN and for LED +. That mix-up is a dead Pi.

Length and noise

- GPIO A/B and button NO **under ~40 cm**.
- Twist each encoder pair. Do not tape encoder wires along the 5 V LED bundle or a USB 3 cable.
- Star **GND** at header pins 6 / 9 / 14 — do not daisy LED – through a switch C then back to the Pi.
- LED rings are tens of mA; pin 2 is enough. The official **USB-C 5 V / 5 A** cable is the only high-current wire.
- Sleeve the run in PET braid. Hot-glue strain relief at the panel.

Must: no 5 V on BCM; PCIe flex seated, no crease; clamp USB-C + HDMI0 (no USB-C extension); PETG vents above cooler / below Duo; strain-relieve to the case. One kiosk: ferrule + Faston + braid. Terminal on a 40-pin riser if it will not fit. Official Pi 512 GB 2230 in Duo slot A. Panel USB 3.0 27 mm to Pi USB 3; P047 on USB 2.